Fix The Exhibit

User Manual

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1. This Document

Fix the Exhibit is a browser-based game created to run on a museum kiosk. This document serves as the help and handover guide for the project. It was written assuming the reader has not seen the project before and has no programming background for standard daily operations. If a task ever requires changing a file, this guide points out the exact file name and keywords to search for, along with the specific edit required to be made. The contents are split into different sections based on how various types of people will interact with the game.

Table 1.1. *Section Readers*

Reader	Purpose	Section Number
Visitor	Player of the game at the kiosk.	2
Museum Staff	Responsible for opening the game each morning, keeping the kiosk running smoothly throughout the day, resetting the session when needed, and helping visitors with any questions.	2, 3 and 4
Content Managers	Manages content changes, including updating artefact information, facts, quiz questions, graphics, and adding new game levels.	5 and 6
Technical Admin	Handles game hosting, modifies configuration settings in the code, and fixes bugs or system errors as they come up.	5, 6, 7, 8, and 9

Table 1.2. *Game Location*

Item	Link/Locations	Description
Live Game	https://mq-playthroughhistory-2026.github.io/FixTheExhibit/	Open this link in any browser to launch the game. It runs directly off the repository's main branch.
Source Code and Content	https://github.com/MQ-PlayThroughHistory-2026/FixTheExhibit	Contains all necessary game files, including the content scripts that staff can edit.
GitHub Repository Guides	README.md, CONTRIBUTING.md and data/levels/README.md in the repository.	Brief technical documentation written specifically for developers.

2. Playing the Game

The player is the new intern at a museum. A delivery arrives with labels mixed up, so real artefacts from the exhibit's period are mixed with packages that don't belong. Each level has three stages. In the sorting stage, the player drags packages off a moving conveyor belt into the display case or into the storage vault before the timer runs out. In the arranging stage, the player places the real artefacts in the display case and reads a short description of each. In the quiz, the player answers one question per artefact. A certificate then gives a score out of 100 and a title, from Curious Visitor up to Master Historian. The whole level takes about eight to twelve minutes.

The game guides visitors as they play. Pressing “Start” for the first time opens an introductory instruction card, and a brief “How to Play” prompt appears at the start of each new stage. Since the game does not store any user data, every session resets automatically for the next player. Most cards and other

popups have a speaker icon in the top right corner, which will start reading the card aloud when tapped. Tapping the icon again or exiting the card will stop the reading.

2.1. Starting the Game

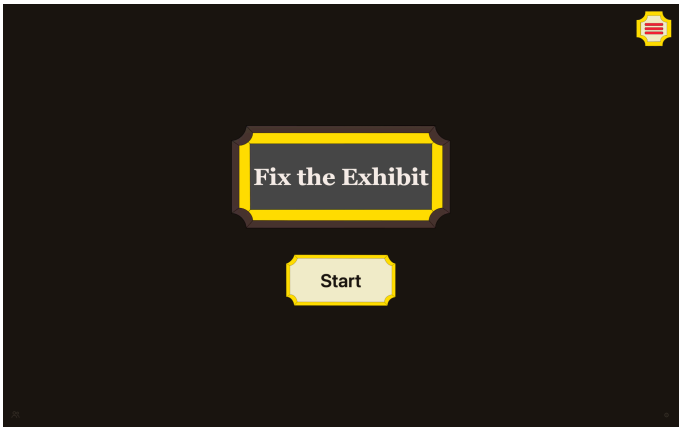


Figure 2.1.1. *Main Menu*

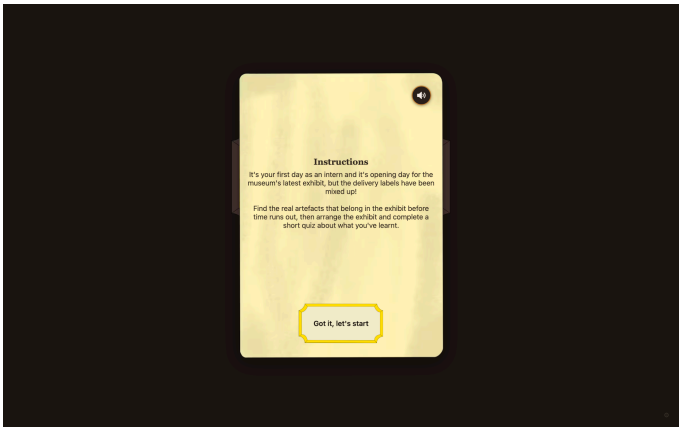


Figure 2.1.2. *Instructions Card*

1. Tap the screen to begin. On a kiosk, this expands the game to full screen and starts the background music.
2. Start the game. Press Start, read through the introductory instructions, and tap "Got it, let's start".
3. Select a level. Choose between two available options: Australian Gold Rush or Ancient Egypt.
4. Select difficulty.
    a. Easy: Features a 90-second sorting timer, a slower conveyor belt, silhouettes during the arranging stage, and easier quiz questions.
    b. Hard: Features a 45-second sorting timer, a faster conveyor belt, clue cards during the arranging stage, and harder quiz questions.



Figure 2.1.3. *Level Selection*



Figure 2.1.4. *Difficulty Selection*

2.2. Sorting Stage

Packages move along the conveyor belt from left to right, looping back around if they are not picked up immediately. Drag each package up into the display case if it matches the current historical period, or drop it into the storage vault if it does not. The top progress bar fills up as artefacts are sorted, while the timer counts down. The timer allows four seconds per package plus 10 on easy, and two per package plus five on hard, so it automatically adjusts based on the number of packages. Sorting an item correctly plays a chime, while an incorrect drop plays a buzz. Both actions trigger an information card with a fun fact, so mistakes still offer a learning opportunity. The conveyor belt and timer both pause whenever a fact card is open.

Once all packages are sorted, or when the time runs out, a “Well Done” or “Time’s Up” screen appears. This screen displays the player's final score and gives them three options: *Next* to proceed, *Restart* to try again, or *Leave* to return to the main menu. Players who time out do not get the *Next* option.

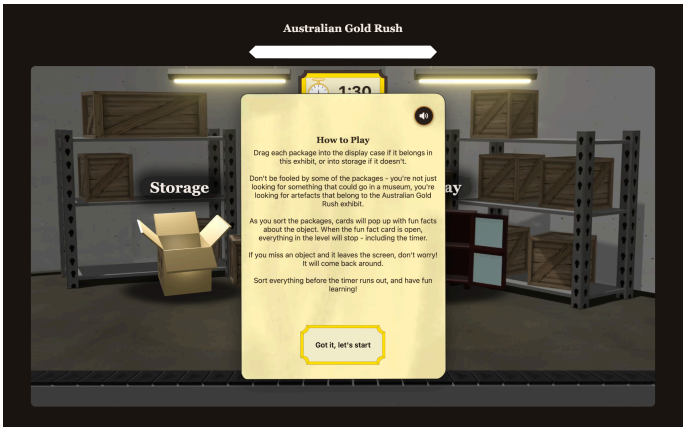


Figure 2.2.1. *Sorting - How to Play*

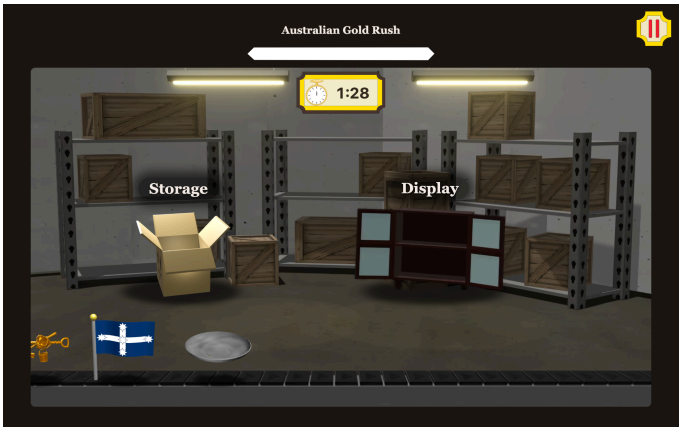


Figure 2.2.2. *Sorting Stage*

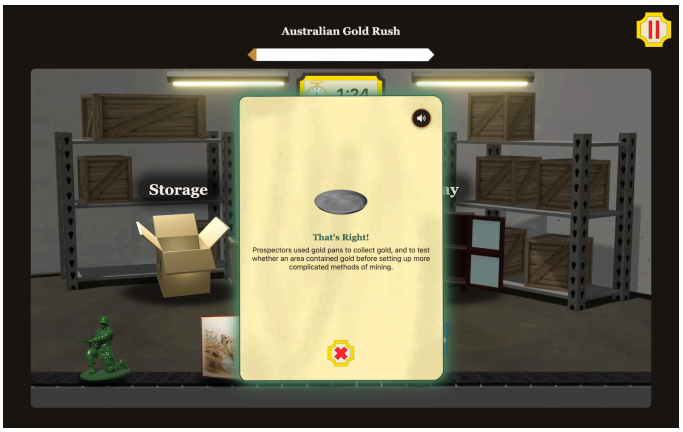


Figure 2.2.3. *Correct Sort Card*



Figure 2.2.4. *Wrong Sort Card*

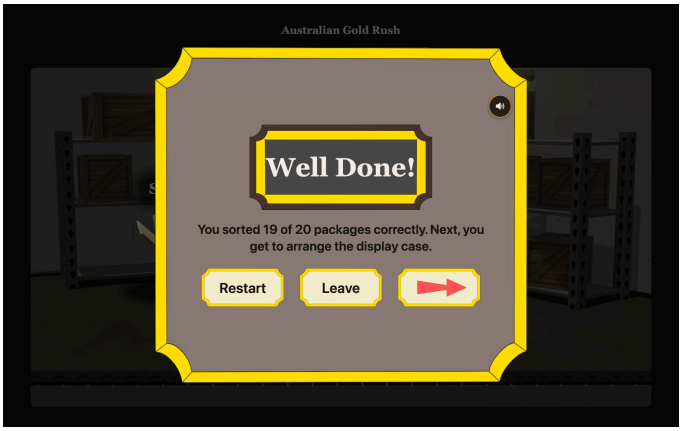


Figure 2.2.5. *Sorting Well Done*

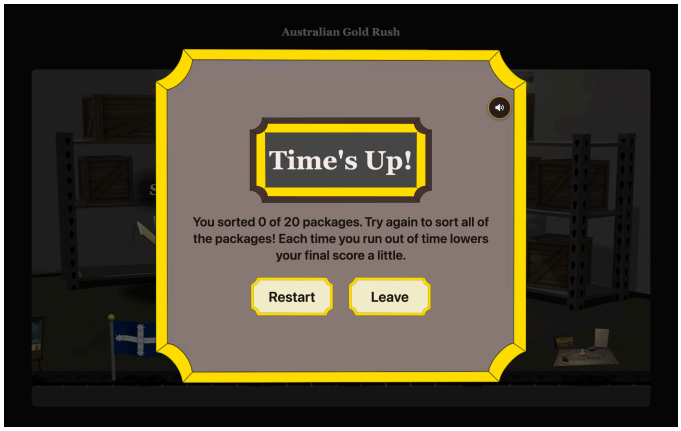


Figure 2.2.6. *Time's Up Panel*

2.3. Arranging Stage

The real artefacts sorted in the previous stage now sit beside the display case, waiting to be dragged onto the shelves. On Easy mode, each shelf spot displays the artefact's silhouette. On Hard mode, spots display clue cards instead, which players can tap for a clue.

Correctly placing an artefact locks it in and opens its information card. This card works like a mini booklet: the first page features an image and summary, while following pages offer deeper context

and a QR code to sources (navigated using the corner arrows or clicking on the next card). The information cards can be reopened at any time if you accidentally click away to close them. Incorrect placements flash briefly before the artefact slides back into the tray. There's no timer for this stage, so visitors can read at their own pace, though incorrect drops will reduce their final score.

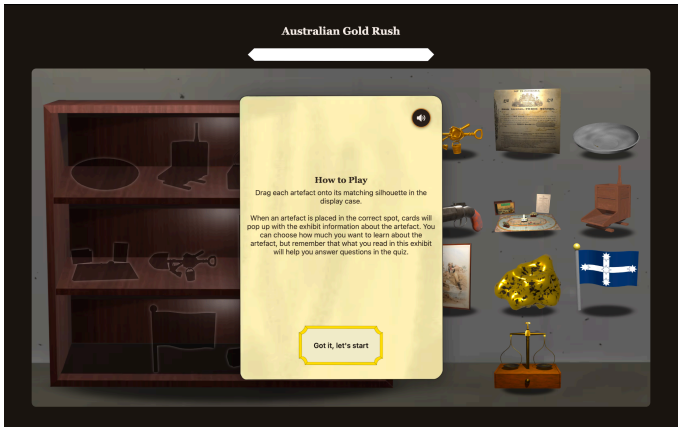


Figure 2.3.1. *Easy Arranging Instructions*

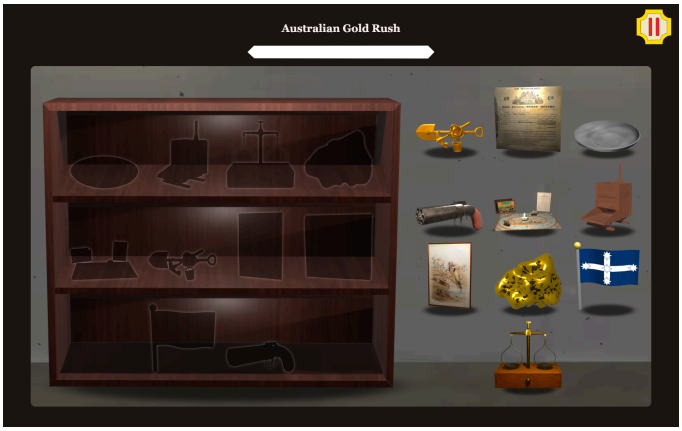


Figure 2.3.2. *Arranging on Easy*

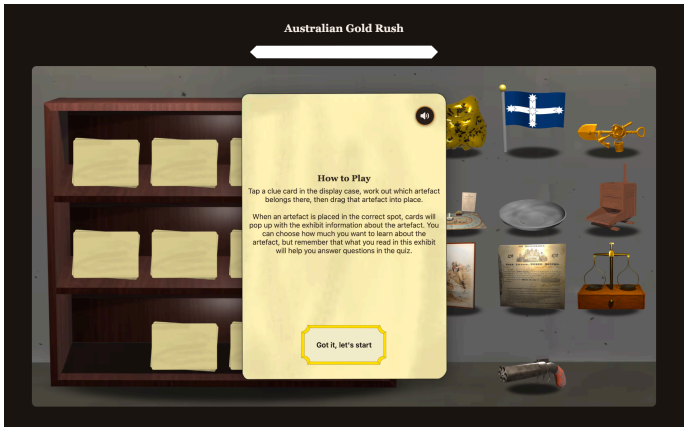


Figure 2.3.3. *Hard Arranging Instructions*

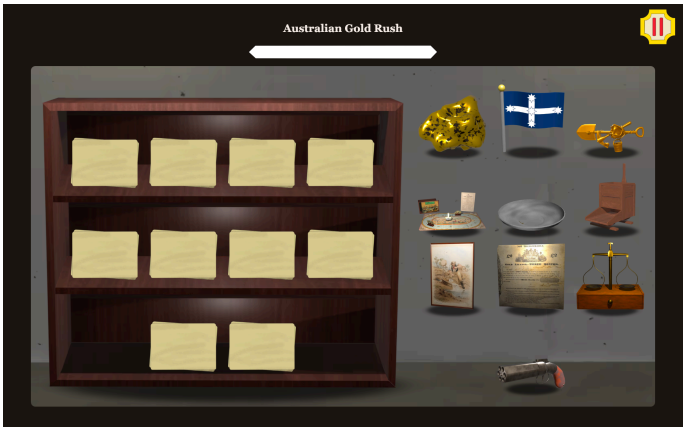


Figure 2.3.4. *Arranging on Hard*



Figure 2.3.5. *Clue Popup*

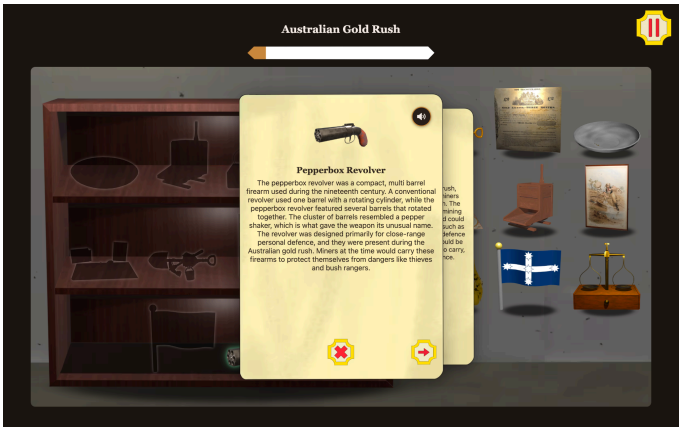


Figure 2.3.6. *Information Card*

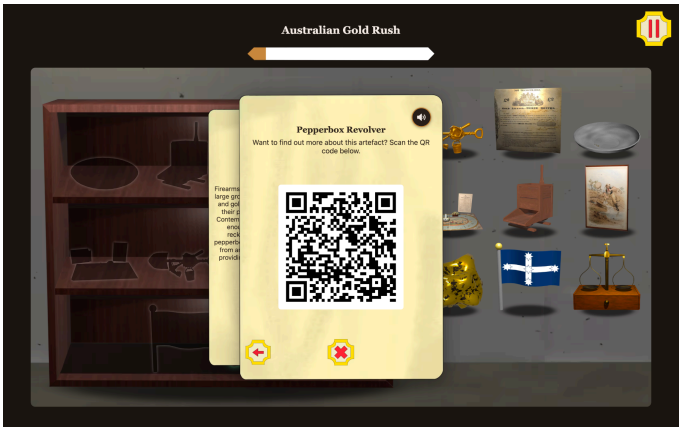


Figure 2.3.7. *Source QR Code Card*

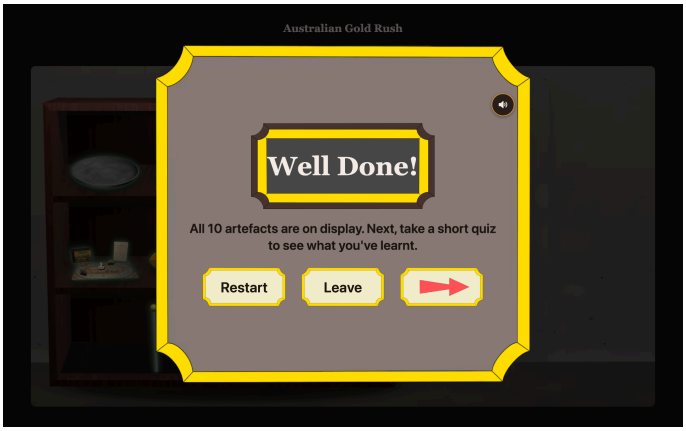


Figure 2.3.8. *Arranging Well Done*

2.4. Quiz Stage

Once the final artefact is placed, tapping Next starts the quiz. Each artefact in the level has one multiple-choice question with four options (A to D). Tapping an answer selects it instantly, and correct choices turn green, while incorrect ones turn red, so visitors always know the right answer before clicking the arrow to continue.

After the last question, a summary panel displays the final quiz score, and a prompt opens the completion certificate. This certificate congratulates the player, displays the exhibit name, and awards a title alongside a score out of 100. From there, players can select Restart to replay the level or Leave to return to the title screen. Table 2.1 details how scores and titles are calculated. Note that scores are displayed purely for visitor feedback and are never saved.

Table 2.4.1. *Score and Title Calculations*

Scene or Title	Rule
Sorting - 30 Points	Percentage of packages placed in the correct zone. If the player has timed out, this will be scaled depending on the number of time-outs.
Arranging - 20 Points	Ratio of correctly placed artefacts to total drop attempts. Trying multiple spots to guess the right position reduces this score.
Quiz - 50 Points	Percentage of questions answered correctly out of the total.
Master Historian	85+ on Easy | 90+ on Hard
Museum Curator	70-84 on Easy | 75-89 on Hard
Artefact Investigator	50-69 on Easy | 55-74 on Hard
History Explorer	30-49 on Easy | 35-54 on Hard
Curious Visitor	Below 30 on Easy | Below 35 on Hard



Figure 2.4.1. *Quiz Question*



Figure 2.4.2. *Quiz Answered Correctly*



Figure 2.4.3. *Quiz Answered Incorrectly*

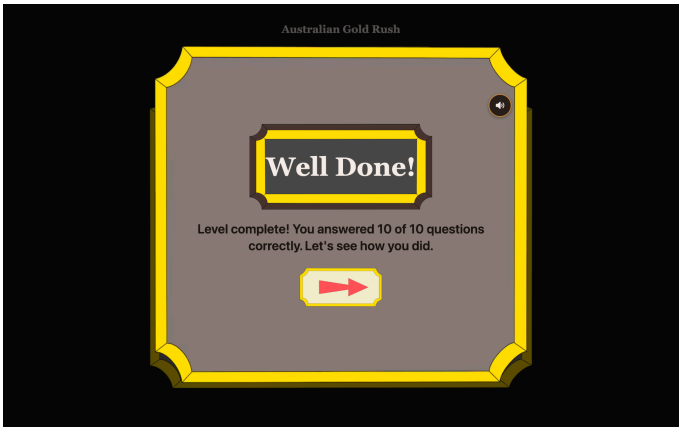


Figure 2.4.4. *Quiz Well Done*

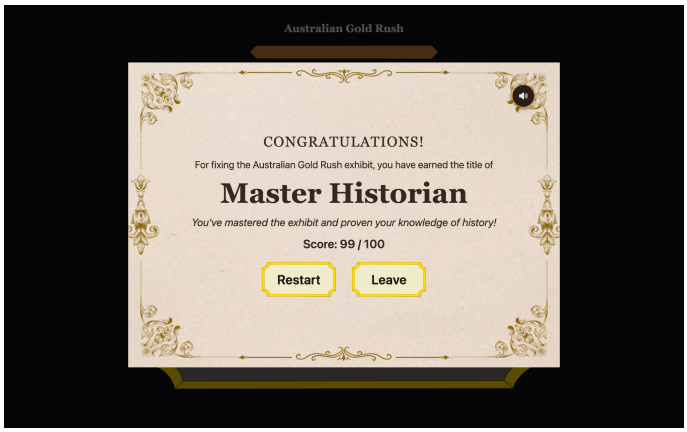


Figure 2.4.5. *Certificate*

2.5. Pause/Settings Menu

Tapping the menu button at the top right of the menu scenes opens the audio settings menu. Players can also press Escape on a keyboard to toggle the menu open or closed. The settings menu includes volume sliders and mute toggles for background music, sound effects, and text-to-speech sound. After changing the volume on sound effect or text-to-speech sliders, a short sound effect or text-to-speech sample will be played. During gameplay, pressing Escape on a keyboard or tapping the pause button at the top right of the screen will open the pause menu. This contains all audio settings, alongside Restart (restarts the current stage), Resume, and Quit (returns to the title screen for the next player) buttons. Pressing the Restart and Quit buttons will open a confirmation message to ensure players want to take that action, as both actions will result in a loss of game progress.

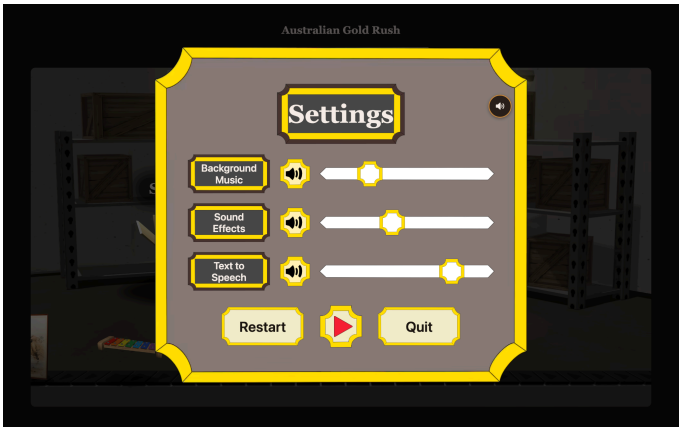


Figure 2.5.1. *Pause Menu*

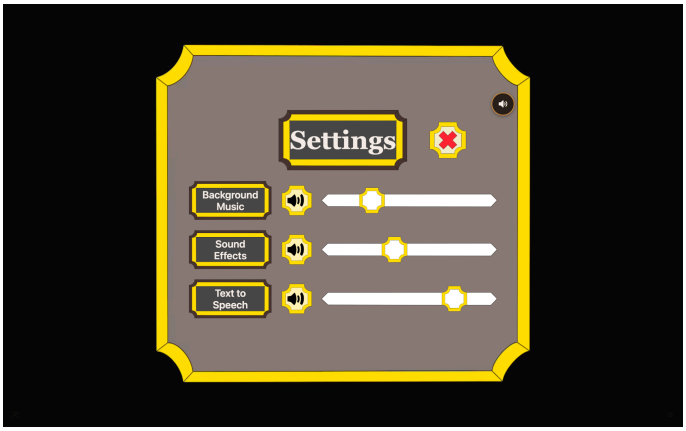


Figure 2.5.2. *Sound Settings*



Figure 2.5.3. *Leave Game Confirmation*



Figure 2.5.4. *Restart Stage Confirmation*

3. Museum Staff - Kiosk Mode

This section covers the day-to-day running of the kiosk.

3.1. Opening the Game Each Morning

1. Turn on the kiosk and check that it is connected to the internet (not needed if the game runs from a local copy, see Section 4.5).
2. Open Google Chrome or Microsoft Edge and go to the live game link in Table 1.2, or double-click start-game.bat if the kiosk launcher has been set up (see Section 4.4).
3. Check that the title screen appears. Tap **Start**, confirm both levels are listed, then reload the page (**F5**) so the first visitor sees the instruction card (tutorial).
4. Tap the screen once. The game switches to fullscreen on the first tap, because browsers do not allow a page to go fullscreen by itself (you can skip this with Section 4.4's kiosk launcher).
5. Check that music plays (it starts after the first tap) and set the kiosk's own volume to a comfortable level or adjust the default settings in Table 6.1.

3.2. Automatic Reset After Inactivity

If nobody touches the screen for **3 minutes**, an **Are you still there?** message appears with a **60-second countdown**. Any touch, or the **Continue** button, closes it and play carries on. If the countdown reaches zero, the page reloads to the title screen, clearing the previous visitor's progress and volume settings. In a normal browser, reloading may leave fullscreen (which will allow access to the OS),

and the next visitor's first tap turns it back on; in kiosk mode (Section 4.4), the screen stays fullscreen. The timings can be changed, see Table 6.1.

3.3. Resetting a Session Manually

To give a new visitor a clean start at any time, use **Pause → Quit** during a stage, **Leave** on any pop-up or the certificate, or press **F5** on a connected keyboard. Each of these reloads the game to the title screen.

3.4. Exiting Fullscreen (Staff Only)

1. On the title screen, tap the faint gear icon (⚙) in the bottom-right corner. It is deliberately hard to see so visitors leave it alone.
2. Type the staff passphrase and tap **Confirm**. A wrong passphrase shows "Incorrect passphrase." **Cancel, Escape** or tapping outside the box closes it.
3. To change the passphrase, see Table 6.1.

Important: the passphrase is set to the development placeholder exit and must be changed before the kiosk is open to the public (Table 6.1). It only deters visitors; it is not real security. When you start the game with a kiosk launcher (highly recommended for real deployment, as it provides real security; (see Section 4.4), the gear icon is removed, and staff exit becomes **Alt + F4**. iPad Safari does not support fullscreen, so use Guided Access there.

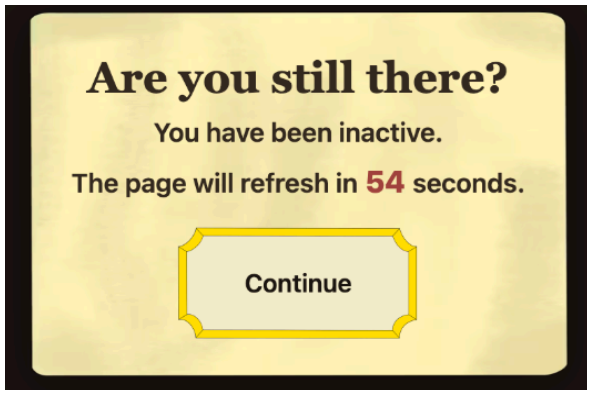


Figure 3.4.1. *Inactivity Warning*

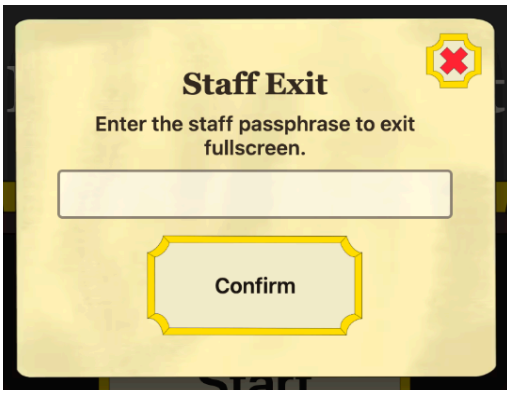


Figure 3.4.2. *Staff Exit Passphrase Box*

3.5. Closing at the End of the Day

Exit fullscreen (see Section 3.4 or use the kiosk's own exit), close the browser and shut the device down. The game stores no visitor data, so there is nothing to save or clear.

3.6. Helping Visitors

Table 3.6.1. *Common Visitor Questions*

The Visitor May Say…	Suggested Answer
"The timer ran out."	Tap the arrow to keep going, or Restart to try sorting again. Unsorted packages only lower the player's score.
"I can't tell where an artefact goes."	In Hard mode, tap a clue card to read its clue. Easy mode shows silhouettes instead.
"Nothing happens when I drag."	A card or menu is open. Close it with the red X or the Resume arrow.

“There’s no sound.”	Open the pause or settings menu and check that the sliders are not at zero or muted, then check the kiosk volume.
“The speaker button doesn’t do anything.”	If other sounds are working, open the pause or settings menu and check that the Text to Speech slider is not at zero or muted.
“The game went back to the start.”	It resets after about four minutes without a touch so the next visitor starts fresh.
“Can I save my score?”	No. Results are not stored, but visitors are welcome to photograph their certificate.

4. Installing and Hosting

Fix the Exhibit is a set of static files (HTML, CSS, JavaScript, JSON, images and audio). It needs no database, no server software of its own, and no installation on the visitor’s side. It only needs to be served to a web browser.

4.1. System Requirements

Table 4.1.1. *System Requirements*

Item	Requirement
Browser	Google Chrome or Microsoft Edge (recommended). Firefox and desktop Safari also run the game. iPad Safari runs it but cannot go fullscreen.
Screen	Designed for a 1920 x 1080 touchscreen, but the layout scales down to laptops, tablets and phones.
Input	Touch or mouse. A keyboard is optional and only useful to staff.
Internet	Required when hosted on GitHub Pages; not required for the local fallback (see Section 4.5).
Hosting	GitHub Pages (recommended, free) or a local copy served by a small local web server.

4.2. Getting Your Own Copy

The original repository stays under the project team’s GitHub organisation, and the team provides no support after handover. The museum should therefore take its own copy, using either option below:

Option A - Fork on GitHub (recommended). Requires a free GitHub account. (GitHub, n.d.-c)

1. Sign in to GitHub and open the source code link in Table 1.2.
2. Click **Fork** near the top-right of the page.
3. Choose the museum’s account as the owner, keep the name “*FixTheExhibit”* and click **Create fork**. The museum now owns a full copy at github.com/<account>/*FixTheExhibit*.

Option B - Download a ZIP. No GitHub account needed. On the repository page, click the green **Code** button, then **Download ZIP**, and unzip it. The ZIP archive supplied at handover can be used the same way. A ZIP copy cannot use GitHub Pages, so you must host it another way (see Section 4.5).

4.3. Hosting on GitHub Pages (Recommended)

1. In the museum’s fork, open **Settings**, scroll down to **Pages**.

2. Under **Build and deployment**, set **Source** to **Deploy from a branch**, choose the branch main and the folder / (root), then click **Save**.
3. After one or two minutes, the page shows the game's address in the form https://<account>.github.io/FixTheExhibit/. Open it on the kiosk and set it as the browser's home page.

Every change committed to the main branch republishes automatically within a few minutes, updating every kiosk at once. The repository's **Actions** tab shows each "pages build and deployment" run; a green tick means the change is live. GitHub Pages is free, put only for public repositories. GitHub (n.d.-a)

4.4. Starting the Game in Kiosk Mode (Recommended for Museums - More Security)

On Windows: the simplest way to lock the game on a screen is a small launcher file, start-game.bat, that opens Google Chrome in kiosk mode: fullscreen, with no address bar, tabs, or menus for visitors to reach. Kiosk mode also stays fullscreen through the inactivity reset. The launcher adds ?kiosk to the end of the game's address, which tells the game to remove the in-game Staff Exit. This is important: typing into the Staff Exit's password box makes Chrome offer to save the password, and that pop-up could let a visitor access the OS.
To start the game in kiosk mode:

1. Open Notepad and paste the lines below, replacing the GAME_URL address with the museum's own game link (Section 4.3) and append ?kiosk.
2. Save it as start-game.bat (set **Save as type** to **All Files**), e.g., to C:\FixTheExhibit.
3. Double-click start-game.bat to start the game. Staff exit kiosk mode with **Alt + F4**.
4. To start the game automatically whenever the kiosk is turned on, press **Windows + R**, type shell:startup, and place a shortcut to start-game.bat in the folder that opens.

```
@echo off
set GAME_URL=https://mq-playthroughhistory-2026.github.io/FixTheExhibit/?kiosk
start "" chrome --kiosk --no-first-run --user-data-dir="%LOCALAPPDATA%\FixTheExhibitKiosk" "%GAME_URL%"
```

The start-game.bat provided in the handover is linked to the development team's own GitHub Pages link, replace it with your own to ensure longevity (Section 4.3). The separate --user-data-dir profile makes Chrome open a new kiosk window even if Chrome is already running. Note: doing so will create a new folder in C:\Users\<user>\AppData\Local\FixTheExhibitKiosk. To use Microsoft Edge instead, replace chrome --kiosk with msedge --edge-kiosk-type=fullscreen.

4.5. Fallback: Running From a Local Copy

Use this only if the venue cannot rely on internet access. Double-clicking index.html will not work, because browsers block a page opened from a file from loading the level data. A small local web server is needed instead, and developers can use npx serve . , as described in the repository's README. (Windows OS steps shown):

1. Install Python 3 from python.org, ticking **Add python.exe to PATH** during installation.
2. Copy the game folder to C:\FixTheExhibit.
3. Create start-game.bat as in Section 4.4, but with the lines below, which first start the server (minimised) and then open the game from it in kiosk mode.

```
@echo off
cd /d C:\FixTheExhibit
start "Fix the Exhibit server" /min python -m http.server 8000 timeout /t 3 >nul
set GAME_URL=http://localhost:8000/?kiosk
start "" chrome --kiosk --no-first-run --user-data-dir="%LOCALAPPDATA%\FixTheExhibitKiosk"
"%GAME_URL%"
```

Do not close the minimised server window. If the browser later says “This site can’t be reached”, the server has stopped; exit with **Alt + F4** and run start-game.bat again. When content changes, the updated files must be copied onto every kiosk by hand.

4.6. Other Lock-Down Options

- **Windows:** for a dedicated device, Assigned Access can run Microsoft Edge as the only app in kiosk mode, so staff and visitors cannot reach the desktop at all (Microsoft, 2026).
- **iPad:** turn on Guided Access under **Settings → Accessibility → Guided Access**, open the game in Safari and triple-click the top (or Home button) to start or end it. (Apple, 2026).

The in-game Staff Exit (see Section 3.4) is only intended for a normal browser window. When a kiosk browser is used, open the game’s address with a ?kiosk at the end so the Staff Exit is removed.

5. Editing Game Content

All text, images, and quiz questions for a level are stored in dedicated content files, kept completely separate from the underlying game code. Updating a fun fact, swapping an image, or adding a new level simply means editing these files, the game loads the updated content automatically the next time it runs. While you do not need programming knowledge, the files must follow a specific structure. The following sections break down each component and walk through a complete example.

5.1. Content Locations

Table 5.1.1. *Content Files and Folders*

File/Folder	Contains
data/levels/index.json	Lists each available level with its ID, title, and short description.
data/levels/<level-id>/artefacts.json	Holds all data for a level's ten real items (names, facts, descriptions, clues, and quiz content).
data/levels/<level-id>/fillers.json	Holds names and facts for the ten extra packages used during sorting.
assets/<level-id>/artefacts/ and assets/<level-id>/fillers/	Image files linked to each artefact and filler package.
assets/sounds/	Background music and the nine main sound effects.
assets/ui/	UI graphics like buttons and cards. Typically not edited, only added.
index.html	General text used across the app, including instructions and difficulty details.

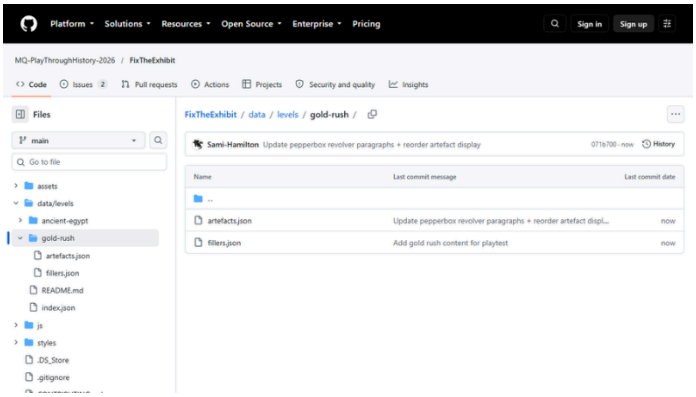


Figure 5.1.1. *GitHub Level Folder*

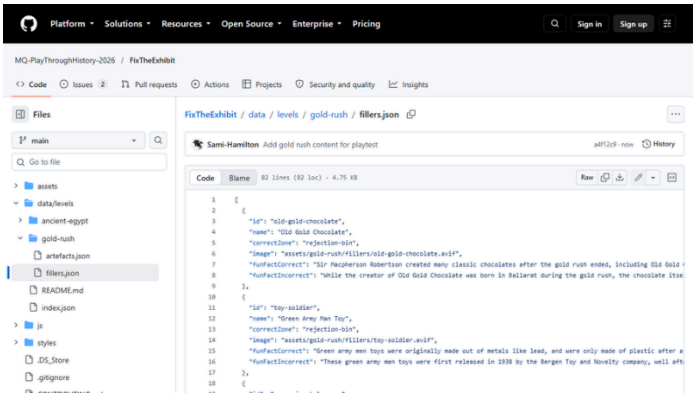


Figure 5.1.2. *GitHub File Editor*

5.2. Editing GitHub Files

Updating files on GitHub requires no extra software, and any saved changes publish to every kiosk automatically. You will need a GitHub account with write access to the project repository. GitHub (n.d.-b)

1. Sign in to GitHub and navigate to the repository.
2. Open the target folder path to reach your file (for example: data > levels > gold-rush > fillers.json).
3. Click the pencil icon near the top right of the page to open the file editor.
4. Make your edits. Be sure to keep all quotation marks, commas, and brackets intact, changing only the text inside the quotes.
5. Click “Commit changes…” at the top right.
6. In the pop-up window, add a brief description of your update (for example: "Fix typo in Old Gold Chocolate fun fact"). Ensure "Commit directly to the main branch" is selected, then click Commit changes.
7. Wait one to two minutes and refresh the game. Your changes are now live. If the updates do not appear, refer to Section 8.

Note: Committing directly to the main branch publishes updates immediately. For major updates like adding a new level, it is safer to draft changes in a separate branch or fork and test them locally first (see Section 4.5).

5.3. Artefact and Filler Data Fields

Both files contain a list of entries. Each individual entry is enclosed in curly brackets, entries are separated by commas, and the entire list is wrapped in square brackets. Table 5.3.1 outlines every field available for an artefact entry, while Table 5.3.2 covers the shorter filler entry format.

Table 5.3.1. *Artefacts.json Fields*

Field	Description
id	A short, unique identifier written in lowercase with hyphens (for example, “miners-cradle”). It stays fixed once created and is never visible to visitors.
name	The title displayed on the package, placement spot, and information cards.
correctZone	Always set to “display-case” for real artefacts.

image	The file path to the image asset (for example, assets/gold-rush/artefacts/miners-cradle.avif). If the image is missing or the path is incorrect, the game falls back to showing a plain box with the item's name.
blurb	A short summary of roughly 150 words that appears on the card after a correct placement.
longExplanation	A list of longer background paragraphs and sources intended for detailed reading. Stored in the data structure for a planned "read more" feature.
funFactCorrect	One or two sentences displayed when the package is correctly sorted into the display case.
funFactIncorrect	One or two sentences displayed if the package is mistakenly sorted into storage.
clue	The clue shown during the arranging stage on Hard mode. If omitted, it defaults to "No clue written yet".
quiz	Two questions (one for Easy mode and one for Hard mode). Each entry includes the question text, exactly four answer choices, and the index number of the correct option (starting at 0 for the first choice). Incomplete or missing entries automatically fall back to placeholder questions.

Table 5.3.2. *Fillers.json Fields*

Field	Description
id	A short, unique name used to identify the artefact.
name	The display name shown directly on the package.
correctZone	Always set to “storage-vault” for filler items.
image	The file path for the item picture (optional).
funFactCorrect	The message shown when the filler item is successfully placed into storage.
funFactIncorrect	The message shown when the item is mistakenly placed into the display case.

5.4. Adding an Artefact (Example)

To add as an example, a miner's cradle to the Gold Rush level:

1. Save the image file to assets/gold-rush/artefacts/miners-cradle.avif.
2. Open data/levels/gold-rush/artefacts.json and scroll to the bottom.
3. Add a comma after the closing curly bracket of the previous entry, then paste the new entry snippet.

Note: Everything is adjusted based on the number of artefacts. This includes the display cabinet, conveyor belt, sorting timer and number of quiz questions.

Table 5.4.1. *Example Code Block*

```
{
  "id": "miners-cradle",
  "name": "Miner's cradle",
  "correctZone": "display-case",
  "image": "assets/gold-rush/artefacts/miners-cradle.avif",
  "blurb": "Miners rocked gravel and water through the cradle so the heavier gold settled behind the riffles
while the lighter waste washed away.",
```

```
  "longExplanation": ["Longer paragraph one.", "Longer paragraph two, with sources."],
  "funFactCorrect": "A cradle could wash far more gravel in a day than a single pan.",
  "funFactIncorrect": "This belongs in the exhibit. Cradles were in use on the Victorian goldfields from 1851.",
  "clue": "I rock like a baby, but I am looking for gold.",
  "quiz": {
    "easy": { "question": "What did miners use a cradle for?", "options": ["Separating gold from gravel",
"Carrying water", "Weighing gold dust", "Melting gold"], "correctIndex": 0 },
    "hard": { "question": "What held the gold back inside a cradle?", "options": ["Riffles", "A sieve", "A
magnet", "Wax"], "correctIndex": 0 }
  }
}
```

Before committing your changes, test the file by launching a local copy of the game and opening the level. If the game immediately returns to the start screen when loading the level, the JSON file could not be parsed. This usually indicates a missing or misplaced bracket or comma.

5.5. Pictures and Sounds

5.5.1. Picture Files

The game uses the AVIF format for images because it offers crisp quality at small file sizes. It also supports WebP, PNG, and JPG formats. Images should be square (roughly 512 by 512 pixels) with a transparent background so the artefact sits neatly on the conveyor belt and renders properly as a silhouette on the display shelf.

5.5.2. File Naming

Name image files using the exact lowercase, hyphenated ID assigned in the JSON entry. Make sure the file path in the image field matches precisely. Paths and file names are case-sensitive on the host server, meaning cradle.avif and Cradle.avif are treated as two different files.

5.5.3. Uploading Images via GitHub

Navigate to the target folder, click “Add file”, followed by “Upload files”, drag your image into the drop zone, and commit the update.

5.5.4. Sound Assets

Background music is located at assets/sounds/background-music/bgm.mp3, while sound effects (e.g. button-click.mp3, correct.mp3, incorrect.mp3) reside in assets/sounds/effects. Swapping any audio file with a new MP3 using the same file name updates the sound in-game without requiring any code edits.

5.5.5. Text-to-Speech

On panels with a speaker button, the text-to-speech functionality will automatically read any text content written on the panel, so the spoken text will automatically update when written text is changed. To add a speaker button to a new panel in index.html, add the “data-tts” attribute to the html element. To add a speaker button to a new panel using JavaScript, call the addTtsButton(<panel>) function, replacing “<panel>” with the name of the new panel. When using the JavaScript function, ensure that you import the function from tts.js using: import { addTtsButton } from '../ui/tts.js'.

5.6. New Level Addition

1. **Select a level ID:** Choose a short identifier, such as “federation”.
2. **Create content files:** Build a new directory at ‘data/levels/federation’ containing ‘artefacts.json’ and ‘fillers.json’, each populated with entries using the required format (ten in each tested for a good balance). Copying the files from ‘gold-rush’ and replacing their content is the fastest way to get started.
3. **Upload image assets:** Create ‘assets/federation/artefacts’ and ‘assets/federation/fillers’, then upload the respective pictures into these folders.
4. **Register the level:** Add an entry to ‘data/levels/index.json’ using the same ID, specifying the title to display on the selection card and a one-line description.
5. **Launch and test:** Reload the game. The new level card will display on screen, allowing players to run through all three stages without requiring any modifications to the core game code.

6. Editing Game Settings

A few settings are stored directly in the code rather than in content files because they adjust gameplay mechanics instead of display text. Each setting occupies a single line in a specific file. You can edit these values following the same process used for content files (see Section 5.2), taking care to change only the specified number or text snippet.

Table 6.1. *Adjustable Settings*

Setting	Location	Change	Description
Easy Sorting Timer	js/game/state.js	sortingTimerSecondsPerObject: 4 and sortingTimerSecondsBuffer: 10	Sets the sorting timer duration based on seconds per package plus a buffer (for example, twenty packages result in 90 seconds on Easy).
Easy Belt Speed	js/game/state.js	beltSpeedPxPerSec: 90	Controls how fast packages move along the conveyor belt in Easy mode. Higher values increase the speed.
Hard Sorting Timer	js/game/state.js	sortingTimerSecondsPerObject: 2 and sortingTimerSecondsBuffer: 5	Sets the sorting timer duration for Hard mode (45 seconds for twenty packages).
Hard Belt Speed	js/game/state.js	beltSpeedPxPerSec: 150	Controls the conveyor belt speed in Hard mode.
Inactivity Time	js/ui/kiosk-refresh.js	IDLE_TIME_LIMIT = 3 * 60 * 1000	The delay in milliseconds without screen interaction before displaying the idle warning (currently set to 180,000 ms, or three minutes).
Warning Countdown	js/ui/kiosk-refresh.js	COUNTDOWN_TIME = 60	The number of seconds the inactivity warning stays on screen before the game automatically reloads.
Staff Passphrase	js/ui/kiosk.js	STAFF_PASSPHRASE = 'exit'	Sets the password string used to restrict administrative access in a standard browser window (bypassed when launching with ?kiosk). Change only the text inside the quotes. Note that this is a basic deterrent rather than strict security,

			as the password can be viewed in the source code.
Default Music Volume	js/ui/audio.js	DEFAULT_BGM_VOLUME = 0.25	Sets the initial background music volume on a scale from 0 (silent) to 1 (maximum volume).
Default Effects Volume	js/ui/audio.js	DEFAULT_SFX_VOLUME = 0.4	Sets the initial sound effects volume on a scale from 0 (silent) to 1 (maximum volume).
Default Text-To-Speech Volume	js/ui/tts.js	DEFAULT_TTS_VOLUME = 0.8	Sets the initial text-to-speech volume on a scale from 0 (silent) to 1 (maximum volume).
Preferred Text-To-Speech Voice	js/ui/tts.js	PREFERRED_LANGS = ['en-AU', 'en-GB', 'en-US', 'en']	Sets the preferred language/voices for text-to-speech functionality. The left-most voice will be used first, but if it is unavailable the program will try using the other voices, moving through the list from left to right.
Score Weights	js/game/score.js	WEIGHTS = { sorting: 30, arranging: 20, quiz: 50 }	Defines the point weight allocated to each of the three stages out of 100 total points. Ensure the three stage values always sum to 100.
Titles And Cut-Offs	js/game/score.js	TITLES = [...]	Defines the five end-game titles, including display text and the minimum score required on Easy and Hard modes. Keep the entries ordered from highest score threshold to lowest.

Display text that is not linked to specific artefacts can also be changed in the code. When changing this text, please note that the text-to-speech functionality reads whatever text is on the card, so editing text in .json files or the index.html file will automatically change what is spoken. The only exception to this is the pause menu which has a unique spoken text set in index.html (see Table 6.2).

Table 6.2. *Interface Texts*

Text	File	Location
Game title, Start button, instructions card	index.html	Located in the “screen-main-menu” section near the top of the file.
Level screen heading and subtitle	index.html	Located in the “screen-level-select section”. Note that level titles and historical periods are loaded directly from data/levels/index.json.
Difficulty cards	index.html	Located in the “screen-difficulty-select” section. Update this text if you modify the timer values listed in Table 6.1.
Display case and storage labels, sorting How to Play card	index.html and js/game/sorting-scene.js	Located in the “screen-sorting” section. The exhibit title on this card is dynamically generated inside “initSortingScene”.
Arranging How to Play cards (Easy and Hard wording)	js/game/arranging-scene.js	Found as two sentences toward the end of the file within “initArrangingScene”.

Card titles ("That's Right!", "Not Quite!") and the Well Done and Time's Up messages	js/game/sorting-scene.js	Located inside the “onSorted” handler, “showSortingComplete”, and “onTimeUp” functions.
Certificate wording ("Congratulations!", "you have earned the title of", "Score")	index.html and js/ui/certificate.js	Located in the “certificate” block near the end of index.html, alongside “showCertificate” for the two dynamically generated lines.
Pause menu labels, staff exit box, inactivity warning	index.html	Located near the end of the file in the “pause-menu”, “staff-exit-modal”, and “timeoutModal” blocks. The spoken text for the pause menu is in “data-tts-text”, which should be updated if the slider labels change.
Quiz feedback lines	js/quiz/quiz-scene.js	Found inside the “choose” function.

7. Troubleshooting

Table 7.1. *Common Problems and Solutions*

Problem	Likely Cause	Solution
Level screen displays "Levels could not be loaded"	Syntax error in "data/levels/index.json" (such as a missing comma or unclosed bracket after an edit).	Open the file on GitHub, verify its structure against Section 5. If needed, revert the change using GitHub History.
Selecting a level jumps directly back to the start screen	Syntax error in that level's "artefacts.json" or "fillers.json", or a file is missing entirely.	Validate both JSON files. Press F12 to open the browser console and view the exact file name and line number causing the error.
An item or placement spot displays a plain text box instead of an image	Incorrect file path, missing image upload, or a letter case mismatch.	Verify that the image field matches the exact file name and extension (".avif") in the assets folder, taking note of capitalisation.
A quiz question displays "Placeholder question about..."	Missing quiz entry or an entry with invalid options (must be exactly four options).	Complete the quiz block using the structure in Section 5.4, providing four answer options and a "correctIndex" value from 0 to 3.
Committed changes do not appear on the kiosk screen	Deployment delay on GitHub Pages, browser caching, or changes were committed to a branch other than "main".	Wait two minutes and refresh the page. Confirm on GitHub that changes were merged to "main".
Game fails to enter full-screen mode	Full-screen mode requires an initial user touch, or the app is running on an iPad (Safari blocks automatic full-screen mode).	Tap the screen once to initialise. On iPad devices, configure Guided Access instead (see Section 4.6).

Game audio is not playing	Audio initialisation requires a user touch, or the user muted the sound via the pause menu.	Tap the screen to enable audio. Open the pause menu to check the volume sliders. Reloading the page resets audio to default levels.
Text-to-speech audio is not playing when the speaker button is pressed, or speaker buttons do not appear as expected	Audio initialisation requires a user touch, or the user muted the text-to-speech sound via the pause menu. Alternatively, the browser or device may not have a built-in voice, or may require an internet connection for the voice to work.	Tap the screen to enable audio. Open the pause menu to check the volume sliders. Reloading the page resets audio to default levels. Test text-to-speech on each kiosk before launching the game to ensure there are no text-to-speech issues specific to that kiosk.
Staff exit gear icon is missing on the main screen	The game was launched using the "?kiosk" parameter, which intentionally hides administrative controls.	Use Alt + F4 to exit the browser. To test administrative controls, launch the URL without the "?kiosk" parameter.
Staff passphrase is not accepted	Incorrect passphrase entered, Caps Lock enabled, or the passphrase was updated in "kiosk.js".	Check "js/ui/kiosk.js" for the active passphrase. Note that the entry is case-sensitive.
Visitors are stuck on the "Time's Up" screen	Players failed to clear all items before the timer expired. The "Time's Up" panel intentionally omits the Next button.	This is by design. To make sorting easier, increase "sortingTimerSecondsBuffer" or "sortingTimerSecondsPerObject" in Table 6.1.
Inactivity warning triggers during active gameplay	Idle timeout threshold is too low for slow readers, or touch inputs are not registering.	Increase the "IDLE_TIME_LIMIT" value in Table 6.1. Tapping anywhere on the screen should dismiss the warning popup.
Local copy stops working after restarting the system	Local development server process is no longer running.	Re-run the command from Section 4.5.
Conveyor belt or display shelf graphics are missing	Asset file in "assets/sort-scene" or "assets/arrange-scene" was renamed or deleted.	Compare local folders against the GitHub repository and restore missing files. Note that basic gameplay functions even without visual assets.

8. Help Resources

The developers do not offer post-handover support. All instructions required to maintain the game are provided in this guide and the three short repository manuals, allowing museum staff or a technical admin to manage the system independently.

Table 8.1. *Sources of Help*

Source	Use
This document	Daily operations, hosting setup, content management, settings adjustments, and troubleshooting.
"README.md" in the repository	High-level project overview, local installation steps, and folder structure.
"data/levels/README.md"	Guidelines on content file structures written for content organisers.
"CONTRIBUTING.md"	Coding conventions, branching strategy, and pull request rules for developers.
GitHub Issues page of the repository	Central hub for tracking bugs, feature requests, and task handovers for technical admins.

Table 8.2. *Known Limitations*

Problem	Solution
Scores	Displayed only on the completion certificate. To suit public kiosk requirements, no score data is saved or sent to a server, meaning there is no leaderboard.
Staff Passphrase	Defined directly in the code (Table 6.1) and visible to anyone inspecting the source code. It serves as a deterrent rather than strict security. On a physical kiosk, security is enforced by launching with "?kiosk" and locking the operating system (see Section 4.6).
Internet	The published game requires an active internet connection. Offline kiosks must run the local copy described in Section 4.5, which requires manual updates whenever content changes.
Content Changes	The game lacks an administrative user interface. Content updates are made by directly editing the files listed in Section 5. Formatting errors in a content file will cause the game to reset to the main screen until fixed (see Section 7).
Level Size	Levels are tested and calibrated for ten artefacts and ten filler items. The game scales automatically to different item counts, using content files to build the cabinet, conveyor belt, sorting timer and quiz. Adding more entries shrinks the display spots and extends stage lengths, so keeping levels close to ten artefacts and fillers allow for an optimal balance.
Text-to-Speech	The text-to-speech functionality uses the browser's or device's built-in voice, so it can differ between kiosks. Additionally, some browsers' voices may need an internet connection, so test this functionality on every real kiosk before launching.

9. References

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